

Tirth Laheri

Data Scientist — Machine Learning Engineer — Data Engineer
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EDUCATION

Middle Tennessee State University

Murfreesboro, TN

Master of Science in Data Science GPA: 4.00/4.00

2024 – Present

- Relevant Coursework: Machine Learning, Statistical Methods, Data Mining, Data Visualization

Parul University

India

Bachelor of Technology in Computer Science (IoT) GPA: 9.06/10.00

2020 – 2024

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Cloud Computing

EXPERIENCE

Graduate Assistant (Administrative)

Aug 2025 – Present

Physician Assistant Department, Middle Tennessee State University

Murfreesboro, TN

- Led the design and rollout of a **SQL-driven data pipeline** to extract, transform, and automate real-time pass-rate and progression metrics for cohorts of **15–20 students**, cutting reporting time by **~90%**.
- Wrote **SQL queries** and Excel-based automation to aggregate attendance, pass rate, and progression data across **3+ cohorts**, and trained department staff on the new system to eliminate manual calculation errors.
- Designed **ETL and data validation workflows** to streamline record intake and integrity checks between the database and reporting layer, presenting recommendations to faculty leadership and reducing administrative processing time by **~40%**.

Data Science Intern

Dec 2023 – Mar 2024

SlashMark

Hyderabad, IN

- Compared **4 classification models** in **Python** and **Scikit-learn** on a **10,000+ record** dataset, improving F1-score by **12%** over the baseline model.
- Conducted exploratory data analysis and feature engineering across **20+ variables**, identifying **5 key predictors** that drove the majority of model performance gains.
- Delivered **weekly analytical reports** and visualizations in **Matplotlib** and **Seaborn** to a 4-person data science team, informing iteration on model selection and feature strategy.

PROJECTS

Multilingual Legal Document Intelligence | *LangChain, OpenAI, pgvector, FastAPI*

Jan 2026 – Present

- Develop a **multilingual legal intelligence platform** supporting **5+ languages** and semantic search across **100+ legal PDF documents** using **OpenAI GPT** and **text-embedding-3-small**.
- Built a **RAG architecture** with **LangChain** and **PostgreSQL pgvector** embeddings, achieving document retrieval latency of **50–500ms**.
- Applied **PaddleOCR** with layout-aware parsing to extract structured information from multilingual legal documents, achieving over **95% OCR accuracy**.
- Exposed scalable **FastAPI** services for OCR, semantic search, contextual Q&A, and document summarization, enabling real-time legal document analysis.

Advanced Agentic RAG System | *Python, LangChain, Multi-Agent, RAG, LLM, Streamlit*

Mar 2024

- Developed a **stateful 10-agent RAG workflow** (Router, Retriever, Grader, Reformulator, WebSearcher, Synthesizer, Generator, FactChecker, SafetyChecker, Clarifier) with conditional routing to dynamically select retrieval, web search, or clarification per query.
- Implemented a **self-correcting retrieval loop** where a Grader agent scores document relevance and automatically triggers query reformulation or live web search via the **Tavily API** on low-confidence retrievals.
- Implemented post-generation **FactChecker and SafetyChecker agents** to verify claims via web search and filter unsafe content, and engineered a **ChromaDB** ingestion pipeline (PDF/DOCX/TXT/URL) behind a configurable **Streamlit** interface.

Diabetic Retinopathy Screening | *Python, TensorFlow, Image Processing, Deep Learning*

Feb 2024

- The Diabetic Retinopathy Screening system utilizes advanced neural networks, including **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)**, for accurate image classification and early detection.
- Achieved high diagnostic accuracy (94.73%) through sophisticated image preprocessing, feature extraction, and deep learning techniques, ensuring sensitivity and specificity in detecting diabetic retinopathy.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++, R, JavaScript, HTML/CSS

Machine Learning & Data Science: Machine Learning, Deep Learning, Statistical Analysis, Computer Vision, NLP, Exploratory Data Analysis (EDA), Feature Engineering

Generative AI & LLMs: LangChain, OpenAI API, Groq, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases (ChromaDB, pgvector)

Frameworks & Libraries: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, Streamlit

Developer Tools & Platforms: Git, VS Code, Google Colaboratory, SQL, PostgreSQL, Power BI, Tableau, Cloud Computing

CERTIFICATIONS

• **Python for Machine Learning & Data Science** - Udemy Certification • **Machine Learning Specialization** - Coursera Certification • **Machine Learning Fast-Track Workshop** - Google Certification • **Cloud Computing** - Parul University • **Introduction to Internet of Things** - NPTEL Certification • **Web Development** - 1Stop.ai

PUBLICATIONS

Visible Light Communication Based Indoor Positioning System 2024

IEEE Xplore Digital Library

- Collaborated with a **4-member research team** to design and develop a VLC-based indoor localization system using photodiode sensors and microcontrollers for accurate indoor navigation.
- Improved positioning precision through light-intensity calibration and ML-driven error correction, validated through joint testing and peer review with co-authors.
- *Published in:* IEEE Xplore Link – ieeexplore.ieee.org/document/10601381

IoT-Based Remote Health Monitoring for Rural Clinics 2023

Private Research Repository, Parul University

- Designed an IoT-based patient monitoring system using **MQTT** for secure real-time vitals transmission (e.g., SpO₂, heart rate) and rule-based anomaly detection to alert remote doctors in rural clinics.

Integration of Smart Wearables and Cloud Computing for Preventive Healthcare 2022

Private Academic Repository, Parul University

- Developed a wearable-cloud architecture with **AI-driven risk detection** to track physiological signals and trigger early health alerts, supporting preventive care and early intervention.